

Program-3: Code a Dockerized Python Flask Application

Step 1: Create Project Directory

```
mkdir Program3  
cd Program3
```

Step 2: Create Dockerfile

Create a file named Dockerfile

```
nano Dockerfile
```

Add the following content:

```
# Use official Python image  
FROM python:latest  
  
# Set working directory  
WORKDIR /app  
  
# Copy requirements file and install dependencies  
COPY requirements.txt .  
RUN pip install --no-cache-dir -r requirements.txt  
  
# Copy application code  
COPY . .  
  
# Expose port 5000  
EXPOSE 5000  
  
# Run the application  
CMD ["python", "app.py"]
```

```
GNU nano 7.2 Dockerfile
FROM python:latest

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 4000

CMD ["python", "app.py"]
```

Save and exit (CTRL+O, ENTER, CTRL+X).

Step 3: Create Flask Application File

nano app.py

Add:

```
from flask import Flask
```

```
app = Flask(__name__)
```

```
@app.route("/")
```

```
def home():
```

```
    return "Hello from Simple Flask Docker!"
```

```
if __name__ == "__main__":
```

```
    app.run(host="0.0.0.0", port=5000)
```

```
GNU nano 7.2 app.py
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from Simple Flask Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0",port=4000)
```

Save and exit.

Step 4: Create requirements.txt

nano requirements.txt

Add:

Flask==2.3.3

Save and exit.

Step 5: Build Docker Image

docker build -t program-3 .

```
1RV24MC109_syeda_ninra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/program3$ docker build -t program3
ERROR: docker: 'docker buildx build' requires 1 argument

Usage: docker buildx build [OPTIONS] PATH | URL | -

Run 'docker buildx build --help' for more information
1RV24MC109_syeda_ninra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/program3$ sudo docker build -t program3 .
[sudo] password for 1RV24MC109_syeda_ninra:
[*] Building 2.8s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 200B
=> [internal] load metadata for docker.io/library/python:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:latest@sha256:97aa8cc0b87a4a312a294d2d4d7b20f6e2a21ed6d4e64ef08c03088c4aa9890f
=> => resolve docker.io/library/python:latest@sha256:97aa8cc0b87a4a312a294d2d4d7b20f6e2a21ed6d4e64ef08c03088c4aa9890f
=> [internal] load build context
=> => transferring context: 461B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:b3a4febd8895184cf66f0c1c689fd7a3b94d346d2a3a250c866866187a5272e9
=> => naming to docker.io/library/program3
```

Verify:

docker images

Step 6: Run Docker Container

docker run -p 5000:5000 program-3

```
=> => naming to docker.io/library/program3
1RV24MC109_syeda_ninra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/program3$ sudo docker run -p 3000:3000 program3
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:4000
* Running on http://172.17.0.2:4000
Press CTRL+C to quit
```

Step 7: Test the Application

Open browser and go to:

http://localhost:5000

Output displayed:

Hello from Simple Flask Docker!

